

A Time For Emoting: When Affect-Sensitivity Is and Isn't Effective at Promoting Deep Learning

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Abstract. We have developed and evaluated an affect-sensitive version of AutoTutor, a dialogue based ITS that simulates human tutors. While the original AutoTutor is sensitive to learners' cognitive states, the affect-sensitive tutor is responsive to their affective states as well. This affective tutor automatically detects learners' boredom, confusion, and frustration by monitoring conversational cues, gross body language, and facial features. The sensed affective states guide the tutor's responses in a manner that helps students regulate their negative emotions. The tutor also synthesizes affect via the verbal content of its responses and the facial expressions and speech of an embodied pedagogical agent. An experiment comparing the affect-sensitive and non-affective tutors indicated that the affective tutor improved learning for low-domain knowledge students, particularly at deeper levels of comprehension. We conclude by discussing the conditions upon which affect-sensitivity is effective, and the conditions when it is not.

Keywords: affect-sensitive, AutoTutor, supportive, confusion, frustration

1 Introduction

Intelligent tutoring systems (ITSs) such as AutoTutor, Andes physics tutor, and Cognitive Tutor, have come a long way towards modeling and responding to learners' cognitive states. This allows ITSs to implement some of the ideal tutoring strategies such as error identification and correction, building on prerequisites, frontier learning (expanding on what the learner already knows), student modeling (inferring what the student knows and having that information guide tutoring), and building coherent explanations [1-7].

However, ITSs can be more than mere cognitive machines, and the link between emotions and learning suggests that they should be affective processors as well [8, 9]. Affect-sensitivity is important for ITSs that aspire to model human tutors because it has been claimed that expert teachers are able to recognize a student's emotional state and respond in an appropriate manner that has a positive impact on the learning process [10, 11]. An affect-sensitive ITS would incorporate assessments of the students' cognitive and affective states into its pedagogical and motivational strategies in order to keep students engaged, boost self-confidence, heighten interest, and presumably maximize learning.

The idea of having a tutoring system detect, respond to, and synthesize affect was once a seductive vision [9]. This vision is now a reality as affect-sensitive learning environments are coming online [7, 12-18]. However, to our knowledge, there has not yet been any evidence demonstrating that, compared to their non-affective counterparts, affect-sensitive ITSs yield impressive learning gains, particularly at deeper levels of comprehension.

We have recently developed an affect-sensitive version of AutoTutor, an ITS with conversational dialogues [19]. The original AutoTutor has a set of fuzzy production rules that are sensitive to the cognitive, but not to the affective states of the learner. The affect-sensitive AutoTutor has a set of production rules that map dynamic assessments of learners' cognitive and affective states with tutor actions to address the presence of boredom, confusion, and frustration. The obvious prediction is that compared to the non-affective tutor, learning gains should be superior for the affect-sensitive version of AutoTutor.

This hypothesis was tested in the current paper where 84 students completed a 60-minute training session with either the regular (non-affective) or the affect-sensitive AutoTutor. Scores from knowledge tests administered before and after each tutorial session were used to determine if the affect-sensitivity yielded enhanced learning.

We begin with a brief description of the affect-sensitive AutoTutor. Since the system is quite complex and a detailed description is beyond the scope of this paper, we focus on a high-level description of the major components of the system. More detailed descriptions can be obtained from previous publications [18, 20-22].

2 The Affect-Sensitive AutoTutor

AutoTutor is a dialogue based ITS for Newtonian physics, computer literacy, and critical thinking. AutoTutor's dialogues are organized around difficult questions and problems (called main questions) that require reasoning and explanations in the answers. When presented with these questions, students typically respond with answers that are only one word to two sentences in length. In order to guide students in their construction of an improved answer, AutoTutor actively monitors learners' knowledge states and engages them in a turn-based dialogue. AutoTutor adaptively manages the tutorial dialogue by providing feedback on the learner's answers (e.g. "good job", "not quite"), pumping the learner for more information (e.g. "What else?"), giving hints (e.g. "What about X?"), prompts (e.g. "X is a type of what?"), correcting misconceptions, answering questions, and summarizing topics.

AutoTutor can keep the dialogue on track because it is always comparing what the student says to anticipated input (i.e., the expectations and misconceptions in the curriculum script). This constitutes AutoTutor's model of the student's knowledge and cognitive states. Pattern matching operations and pattern completion mechanisms drive the comparison. These matching and completion operations are based on symbolic interpretation algorithms [23] and statistical semantic matching algorithms [24].

Boredom, confusion, and frustration are the major negative states that students experience during interactions with AutoTutor and other learning environments [21,

25]. These are states that, if addressed appropriately, can have a positive impact on learning outcomes. Therefore, the current version of the affect-sensitive focuses on detecting and responding to boredom, frustration, and confusion. The tutor is called the *Supportive* tutor because it responds to learners' affective states via empathetic and motivational responses that always attribute the source of the negative affect to itself or the material, instead of the learners' themselves.

2.1 Detecting Affect

The affect detection system monitors conversational cues, gross body language, and facial features to detect boredom, confusion, frustration, and neutral (no affect) (see Figure 1). Automated systems that detect these emotions have been integrated into AutoTutor and have been extensively described and evaluated in previous publications [20-22, 25]. Each channel independently provides its own diagnosis of the student's affective state. These individual diagnoses are combined with a decision-level fusion algorithm that selects a single affective state and a confidence value of the detection. The algorithm relies on a voting rule enhanced with a few simple heuristics.

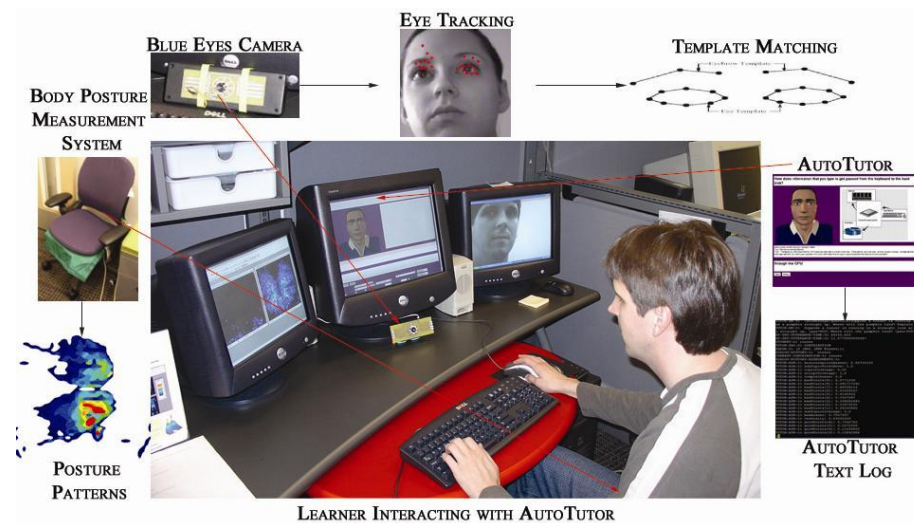


Figure 1. Monitoring affective states during interactions with AutoTutor

2.2 Responding to Affect

An examination of the literature provided some guidance on how best to respond to the states of boredom, confusion, and frustration. We considered some major theoretical perspectives that address the presence of these negative emotions. These included attribution theory [26-28], cognitive disequilibrium during learning [29-32],

politeness [33, 34], and empathy [35, 36]. In addition to theoretical considerations, the assistance of experts in tutoring was enlisted to help create the set of tutor responses.

We created a set of production rules that addressed the presence of these negative affective states by incorporating perspectives from the psychological theories and from the expert's recommendations. When there was no guidance from theory or expertise, the research group added production rules that were intuitively plausible. So in a nutshell, the production rules were determined by theory, experts, and intuition.

The production rules were designed to map dynamic assessments of the students' cognitive and affective states with appropriate tutor actions. There were five parameters in the student model and five parameters in the tutor model. The parameters in the student model included: (a) the current affective state detected, (b) the confidence level of that affect classification, (c) the previous state detected, (d) a global measure of student ability (dynamically updated throughout the session), and (e) the conceptual quality of the student's immediate response. AutoTutor incorporated this five-dimensional assessment of the student and responded with: (a) feedback for the current answer, (b) an affective statement, (c) the next dialogue move, (d) an emotional display on the face of the tutor agent, and (e) an emotional modulation of the voice produced by AutoTutor's text-to-speech engine.



Figure 2. Affective AutoTutor Interface

As an example, consider a student that has been performing well overall (high global ability), but the most recent contribution was not very good (low current

contribution quality). If the current state was classified as boredom, with a high probability, and the previous state was classified as frustration, then AutoTutor might say the following: “Maybe this *topic* is getting old. I’ll help you finish so we can try something new.” This would be a randomly selected phrase from a list that was designed to indirectly address the student’s boredom and to try to shift the topic a bit before the student became disengaged from the learning experience. In this sense, the rules were context sensitive and were dynamically adaptive to each individual learner.

A screenshot of the affective AutoTutor is shown in Figure 2. Here the tutor is displaying a skeptical face because it detected that the student was hedging. Alternate facial expressions include approval, disapproval, enthusiasm, surprise, empathy, and neutral.

3 Method

3.1 Participants and Design

84 participants from a mid south university in the US participated for course credit. The experiment had a between-subjects design where participants were randomly assigned to either the Regular or the Supportive AutoTutor. Participants in each condition completed two training sessions with the *same* version of AutoTutor but on two *different* computer literacy topics (hardware, operating systems, the Internet). Participants did not receive tutoring for the third computer literacy topic; learning gains for this topic were used to assess knowledge transfer. The order in which topics were covered was counterbalanced across participants with a Latin Square.

3.2 Content Covered in AutoTutor Sessions

Participants completed three challenging computer literacy questions in each tutoring session. Hence, each participant received training for six questions over both sessions. Each problem required approximately three to seven sentences of information for a correct answer. The questions required answers that involved inferences and deep reasoning, such as *why*, *how*, *what-if*, *what if not*, and *how is X similar to Y*? Examples of these questions are, “How can John’s computer have a virus but still boot to the point where the operating system starts?” (hardware question), “How would you design an operating system that can manage memory demands from multiple concurrent jobs?” (operating systems), and “How will you design a network that will continue to function, even if some connections are destroyed? (Internet)”.

3.3 Knowledge Tests

Participants were tested on their knowledge of computer literacy topics both before and after the tutorial session (pretest and posttest, respectively). The testing materials were adapted from computer literacy tests used in previous experiments involving

AutoTutor [19]. The items were designed to assess deep levels of knowledge (e.g., “How does the computer assure that other stored information is not overwritten when a save command is given?”) rather than recall of shallow facts (e.g. “What does RAM stand for?”).

Each test had 8 questions on each topic, yielding 24 questions in all. Participants completed alternate test versions for pretest and posttest. The two test versions, composed of different questions, tested learners on the same subject matter and content. The assignment of test versions to pretest versus posttest was counterbalanced across participants.

3.4 Procedure

Participants were tested individually during a 1.5 to 2 hour session. First, participants completed an informed consent and then the pretest. Participants were instructed to take a seat at the computer console and to put on a pair of headphones. Next, the general features of AutoTutor’s dialogue and pedagogical strategies were described to the participant. On the basis of random assignment, participants interacted with either the Supportive or the Regular AutoTutor. They were tutored on one computer literacy topic until three main questions were successfully answered or the 30-minute training period had elapsed (Session 1). They then interacted with the *same* version of AutoTutor on *another* computer literacy topic until three main questions were successfully answered or the 30-minute training period had elapsed (Session 2). Finally, participants completed the posttest and were debriefed.

4 Results and Discussion

Pretest and posttest scores were computed as the proportion of questions answered correctly. Proportional learning gains were computed as $(\text{posttest} - \text{pretest}) / (1 - \text{pretest})$. Separate proportional scores were computed for Session 1 and Session 2. We also computed proportional learning gains for the topic for which participants received no tutoring to assess knowledge transfer.

There was no significant difference in prior knowledge across conditions ($M_{\text{REG}} = .188$, $SD_{\text{REG}} = .090$; $M_{\text{SUP}} = .177$, $SD_{\text{SUP}} = .080$; $p = .557$). Participants were assigned to either a low or a high prior-knowledge group based on a median split on their pretest scores (median = .17). The analyses consisted of 2 (tutor: regular vs. supportive) $\times$ 2 (prior knowledge: low vs. high) between-subjects ANOVAs for learning gains in each session and for knowledge transfer.

Table 1 shows descriptive statistics on proportional learning gains. The main effect for tutor was significant for Session 1, $F(1, 80) = 2.53$, $p = .043$. There was a low to medium sized effect ($d = .375$) in favor of the regular AutoTutor. The tutor $\times$ prior-knowledge interaction was also significant, $F(1, 80) = 3.91$, $p = .051$ (see Figure 3). There was no difference in learning gains across tutors for the low prior-knowledge participants ($d = .017$). However, there was a significant ($p = .022$) and substantial effect ($d = .824$) in favor of the regular tutor for students with high prior-knowledge.

Table 1. Descriptive statistics on proportional learning gains.

Session	Version	Tutor [Main Effect]		Tutor × Prior Knowledge [Interaction]		Prior Knowledge	
		<i>M</i>	<i>SD</i>	Low <i>M</i>	High <i>SD</i>	Low <i>M</i>	High <i>SD</i>
Session 1	Regular	.389	.320	.346	.343	.441	.289
	Supportive	.249	.420	.340	.353	.114	.481
Session 2	Regular	.377	.328	.382	.247	.370	.413
	Supportive	.407	.386	.549	.221	.198	.480
Transfer	Regular	.001	.439	.095	.267	-.113	.572
	Supportive	.092	.387	.244	.244	-.130	.454

There was a different pattern in learning gains for participants' second AutoTutor session. The main effect for tutor was not significant ($p = .973$), but there was a significant tutor × prior-knowledge interaction, $F(1, 80) = 5.07$, $p = .027$ (see Figure 3). Low prior-knowledge participants learned significantly more from the supportive AutoTutor than the regular tutor ($d = .713$). There was no significant difference in learning gains across tutors for the high prior knowledge students.

There was no significant effect for tutor ($p = .448$), nor a significant tutor × prior-knowledge interaction ($p = .338$) for topics for which participants received no tutoring (transfer topic). However, there was a medium sized effect ($d = .583$) in favor of the supportive tutor for the low prior-knowledge students; this effect might be significant with a larger sample.

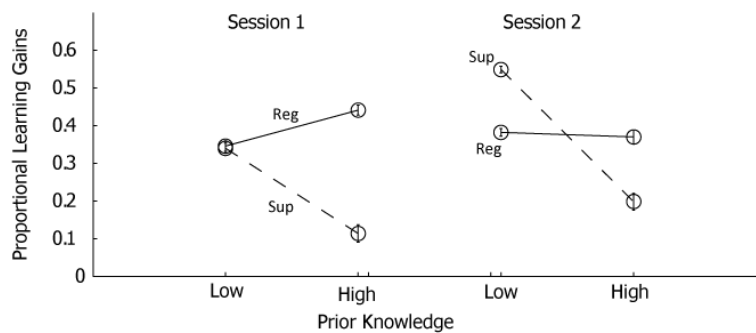


Figure 3. Interactions between prior knowledge and tutor type

Figure 4 portrays the same data in a slightly different way. Low prior-knowledge students show a dramatic improvement ($p < .05$; $d = .71$) in learning gains from Session 1 to Session 2 when interacting with the supportive versus the regular AutoTutor. Learning gains between sessions remained approximately consistent for high prior-knowledge students ($p > .05$).

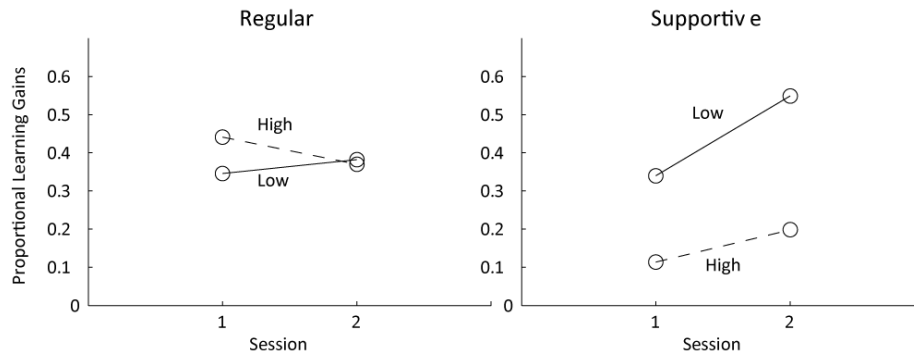


Figure 4. Trends in learning gains across sessions

5 General Discussion

The results of this experiment support a number of conclusions regarding the effectiveness of affect-sensitivity in promoting deep learning gains. First, the supportive AutoTutor was more effective than the regular tutor for low-domain knowledge students in the second session, but not the first session. These results suggest that it is inappropriate for the tutor to be supportive to these students before there has been enough context to show there are problems. Simply put, don't be supportive until the students need support. Second, the students with more knowledge never benefited from the supportive AutoTutor. These students don't need the emotional support, but rather they need to go directly to the content. Third, there are conditions when emotional support is detrimental, if not irritating to the learner. There appears to be a liability to quick support and empathy compared to no affect-sensitivity.

The central message is that there is an appropriate time for affect-sensitivity in the form of supportive dialogues. Just as there is a "time for telling"; there is a "time for emoting." We could imagine a trajectory where low-knowledge students start out with a non-emotional regular tutor until they see there are problems. Then after that they need support, as manifested in Session 2 of our study. Regarding high-knowledge students, they are perfectly fine working on content for an hour or more and may get irritated with an AutoTutor showing compassion, empathy, and care. But later on there may be a time when they want a shake-up AutoTutor for stimulation, challenge, and a playful exchange. Or maybe even a supportive AutoTutor. These are all questions to explore for future research.

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